

	<i>initial net force.</i>	
3(a)	friction on the mass provides the centripetal force for the circular motion of the mass	B1
	For mud to remain on the plate, $0.72W \geq md\omega^2$ $0.72mg \geq 0.35m\omega^2$ $\omega \leq 4.49 \text{ rad s}^{-1}$	M1 C1

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	$\text{number of revolutions} \leq \frac{4.49}{2\pi} \times 60$ $= 42 \text{ min}^{-1}$	A1
	<i>Marker's comments: The question says to explain your working. As such, a statement that friction provides centripetal force to explain before equating the two forces. There were mistakes made in units of ω or to convert rad s^{-1} to number (an integer) of revolutions per minute.</i>	
3(b)	centripetal force increases as r increases/ centripetal force is larger at the edge	M1
	so mud slips off at the edge first.	A1
	<i>Marker's comments: Mud will slip off the plate when friction is <u>not</u> enough to provide the required centripetal force. Students also did not realise that ω is the same for mud near the centre or edge.</i>	
4(a)	acceleration is directly proportional to its displacement from a fixed point	D1